

Advanced (Habanero)

- Q1.** What type of correlation is shown by the data points (1, 2), (2, 3), (3, 5), (4, 7)?
(A) Positive
(B) Negative
(C) None
(D) Cannot be determined
(E) Insufficient data
- Q2.** A line of best fit has a slope of 2 and a y -intercept of -3 . What is the equation of the line?
(A) $y = 2x - 3$
(B) $y = 2x + 3$
(C) $y = -2x - 3$
(D) $y = -2x + 3$
(E) $y = 3x - 2$
- Q3.** The data points (0, 0), (1, 2), (2, 4), (3, 6) show a correlation of
(A) Positive
(B) Negative
(C) None
(D) Cannot be determined
(E) Insufficient data
- Q4.** What is the purpose of a scatter plot?
(A) To show the correlation between two variables
(B) To show the mean of a dataset
(C) To show the median of a dataset
(D) To show the mode of a dataset
(E) To show the range of a dataset
- Q5.** A line of best fit is used to
(A) Show the correlation between two variables
(B) Make predictions based on data trends
(C) Show the mean of a dataset
(D) Show the median of a dataset
(E) Both A and B
- Q6.** The data points (1, 1), (2, 4), (3, 9), (4, 16) show a correlation of
(A) Positive
(B) Negative
(C) None
(D) Cannot be determined
(E) Insufficient data
- Q7.** A scatter plot shows a negative correlation. What can be said about the slope of the line of best fit?
(A) It is positive
(B) It is negative
(C) It is zero
(D) It is undefined
(E) Cannot be determined
- Q8.** The equation of a line of best fit is $y = 2x + 1$. What is the y -intercept?
(A) -1
(B) 1
(C) 2
(D) 3
(E) 4

Open Ended Questions (FRQ)

- Q9.** Using the data points (1, 2), (2, 4), (3, 5), (4, 7), create a scatter plot and draw a line of best fit. Use the line of best fit to predict the value of y when $x = 5$.
- Q10.** Explain the difference between a positive and negative correlation. Provide an example of each.

Chef's Tips

- Tip 1: Ensure students understand the concept of correlation and how to identify it in a scatter plot.
- Tip 2: Emphasize the importance of using the line of best fit to make predictions based on data trends.
- Tip 3: Encourage students to use mathematical notation and terminology when answering questions.